

ECEN 662: Estimation & Detection Theory

Lecture 7: Maximum-Likelihood Estimation

Tie Liu

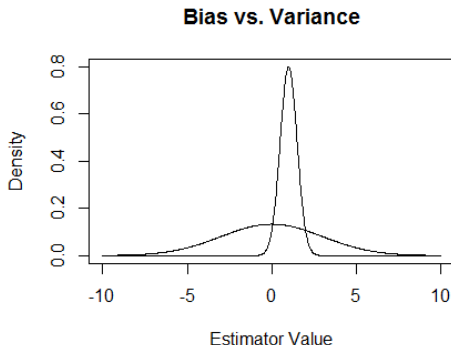
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- More importantly, while an UMVUE minimizes variance among unbiased estimators, strictly adhering to the *unbiased* constraint can sometimes lead to a *significantly* higher variance than a slightly biased estimator



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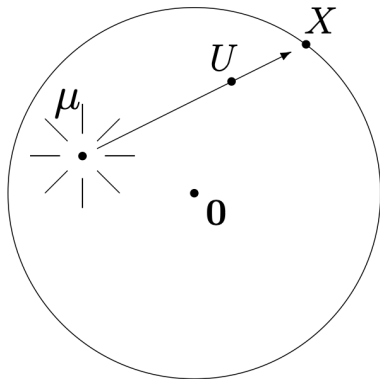
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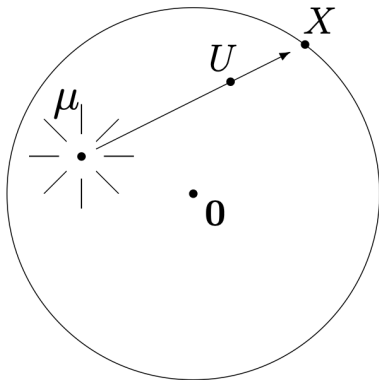
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- Another illuminating example is to estimate the location of a light source based on its projection on the boundary of a dart board over a randomly placed dart





- The details can be found in Michael Hardy's short exposition

Maximum-likelihood estimator (MLE)

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Maximum-likelihood estimator (MLE)

- An alternative *frequentist* estimator is *maximum-likelihood estimator (MLE)*
- Let us first focus on estimating the model parameter θ itself (and defer the discussions on estimating a function of θ to the later part of the lecture)
- Given a likelihood family $f_\theta(x)$, an estimator $\hat{\theta} = \hat{\theta}(X)$ is an *MLE* of θ if

$$\hat{\theta}(x) \in \arg \max_{\theta \in \Theta} f_\theta(x), \quad \forall x \in \mathcal{X}$$

or, equivalently,

$$\hat{\theta}(x) \in \arg \max_{\theta \in \Theta} \log f_\theta(x), \quad \forall x \in \mathcal{X}$$

Maximizing likelihood and minimizing K-L divergence

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- More specifically, given an observation $x = (x_1, \dots, x_n)$, let

$$\hat{p}_x(z) = \frac{1}{n} \sum_{i=1}^n 1_{\{x_i=z\}}, \quad \forall z \in \mathcal{X}$$

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- For two pmfs p and q over the same alphabet $\mathcal{X}$, the *K-L divergence* between p and q is defined as:

$$D_{KL}(p\|q) := \mathbb{E}_p \left[\log \frac{p(Z)}{q(Z)} \right] = \sum_{z \in \mathcal{X}} p(z) \log \frac{p(z)}{q(z)}$$

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- In statistics, information theory, and machine learning, one often use K-L divergence as a *distance* measure between two probability distributions, even though this distance measure is generally *asymmetric* about the two input distributions
- (*Claim*) Maximizing the likelihood

$$\sum_{i=1}^n \log p_{\theta}(x_i)$$

is *equivalent* to minimizing the K-L divergence

$$D_{KL}(\hat{p}_x\|p_{\theta})$$

- First note that

$$\begin{aligned} D_{KL}(\hat{p}_x \| p_\theta) &= \mathbb{E}_{\hat{p}_x} \left[\log \frac{\hat{p}_x(Z)}{p_\theta(Z)} \right] \\ &= \mathbb{E}_{\hat{p}_x} [\log \hat{p}_x(Z)] - \mathbb{E}_{\hat{p}_x} [\log p_\theta(Z)] \end{aligned}$$

Proof

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- Further note that

$$\begin{aligned} \mathbb{E}_{\hat{p}_x} [\log p_\theta(Z)] &= \sum_{z \in \mathcal{X}} \hat{p}_x(z) \log p_\theta(z) \\ &= \frac{1}{n} \sum_{z \in \mathcal{X}} \sum_{i=1}^n 1_{\{x_i=z\}} \log p_\theta(z) \\ &= \frac{1}{n} \sum_{i=1}^n \left(\sum_{z \in \mathcal{X}} 1_{\{x_i=z\}} \log p_\theta(z) \right) \\ &= \frac{1}{n} \sum_{i=1}^n \log p_\theta(x_i) \end{aligned}$$

- Combining the above two results gives

$$D_{KL}(\hat{p}_x \| p_\theta) = \mathbb{E}_{\hat{p}_x} [\log \hat{p}_x(Z)] - \frac{1}{n} \sum_{i=1}^n \log p_\theta(x_i)$$

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- We may thus conclude that minimizing the K-L divergence $D_{KL}(\hat{p}_x \| p_\theta)$ is *equivalent* to maximizing the likelihood $\sum_{i=1}^n \log p_\theta(x_i)$
- The above result also suggests that if our model is *identifiable*, MLE will recover θ in the limit as $n \rightarrow \infty$. We will have more to say about this result a bit later

Re-parameterization

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$$\arg \max_{\theta \in \Theta} f_\theta(x) = \left\{ \theta \in \Theta : \eta(\theta) \in \arg \max_{\eta \in \eta(\Theta)} \tilde{f}_\eta(x) \right\},$$

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- That is, to find an MLE of θ , we can first find an MLE of η and then apply the “inverse mapping” of $\eta(\theta)$

Exponential family of distributions

- Consider an *exponential* family of distribution re-parameterized by the *natural* parameter η :

$$\tilde{f}_{\eta}(x) = h(x) \exp(\eta^t T(x) - A(\eta))$$

where the space of the natural parameter is an *open and connected* subset of $\mathbb{R}^s$

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- To find an MLE for η , note that

$$\nabla_\eta \log \tilde{f}_\eta(x) = T(x) - \nabla_\eta A(\eta) = T(x) - \mathbb{E}_\eta[T(X)]$$

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- A covariance matrix is always positive semi-definite, so for any $x \in \mathcal{X}$, the log-likelihood $\log \tilde{f}_\eta(x)$ is a *concave* function of η , for which any *stationary* point is a global maxima

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- Finally, by the *re-parameterization* property, *any* solution for θ to the equation

$$\mathbb{E}_\theta[T(X)] = T(x)$$

is an MLE for θ

Gaussian with unknown mean and variance

- Recall that in this case, a sufficient statistic is given by

$$T_1 = \sum_{i=1}^n X_i \quad \text{and} \quad T_2 = \sum_{i=1}^n X_i^2$$

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- The above equations have a *unique* solution given by

$$\mu = \frac{T_1(x)}{n} \quad \text{and} \quad \sigma^2 = \frac{T_2(x)}{n} - \left(\frac{T_1(x)}{n} \right)^2$$

- We may thus conclude that the *empirical mean*

$$\hat{\mu} = \frac{T_1(x)}{n} = \frac{1}{n} \sum_{i=1}^n X_i$$

is the MLE for μ and the *empirical variance*

$$\widehat{\sigma^2} = \frac{T_2(x)}{n} - \left(\frac{T_1(x)}{n} \right)^2 = \frac{1}{n} \sum_{i=1}^n X_i^2 - \hat{\mu}^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \hat{\mu})^2$$

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- By *Cochran's theorem*,

$$\mathbb{E}_{\theta} \left[\sum_{i=1}^n (X_i - \hat{\mu})^2 \right] = (n-1)\sigma^2$$

so

$$\mathbb{E}_{\theta} \left[\widehat{\sigma^2} \right] = \frac{n-1}{n} \sigma^2$$

i.e., the empirical variance is a *biased* estimator for the true variance σ^2

- However, compared with the UMVUE

$$\frac{1}{n-1} \sum_{i=1}^n (X_i - \hat{\mu})^2$$

whose variance is given by $2\sigma^4/(n-1)$, the MLE

$$\frac{1}{n} \sum_{i=1}^n (X_i - \hat{\mu})^2$$

has a *smaller* variance $2\sigma^4/n$ (at the expense of being slightly biased)

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- Also note that the variances of the MLE for both μ and σ^2 *coincide* with the CRLB for *unbiased* estimators
- Under certain regularity conditions, it can be shown that the above observations hold for a *general* likelihood function
- More specifically, let $X = (X_1, \dots, X_n)$, where $X_i \stackrel{iid}{\sim} f_\theta(x_i)$. For any θ that is in the *interior* of Θ , let $\hat{\theta}$ be the MLE for θ given by the *unique* solution to the equation

$$\nabla_\theta \left(\sum_{i=1}^n \log f_\theta(x_i) \right) = 0$$

We have $\sqrt{n}(\hat{\theta} - \theta)$ converges in distribution to $\mathcal{N}(0, I_1^{-1}(\theta))$ in the limit as the sample size $n \rightarrow \infty$

- That is, when the sample size n is large, the MLE $\hat{\theta}$ is *approximately* distributed as

$$\mathcal{N}\left(\theta, \frac{1}{n}I_1^{-1}(\theta)\right)$$

or, equivalently,

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- Therefore, under certain regularity conditions, the MLE is *consistent* in that it can *recover* the unknown parameter θ in the limit as the sample size $n \rightarrow \infty$
- The proof of the *asymptotic normality* of the MLE is based on the well-known *central limit theorem*. The details are omitted from this note

Multinomial with unknown probabilities

- Let $X \sim p_\theta(x)$, where the observation

$$X = (N_1, \dots, N_K)$$

is the *empirical counts* over K categories and n total counts, the unknown parameter

$$\theta = (p_1, \dots, p_K)$$

is a *probability* vector over K categories, and the likelihood function is *multinomial*:

$$p_\theta(x) = \frac{n!}{n_1! \cdots n_K!} \prod_{k=1}^K p_k^{n_k}$$

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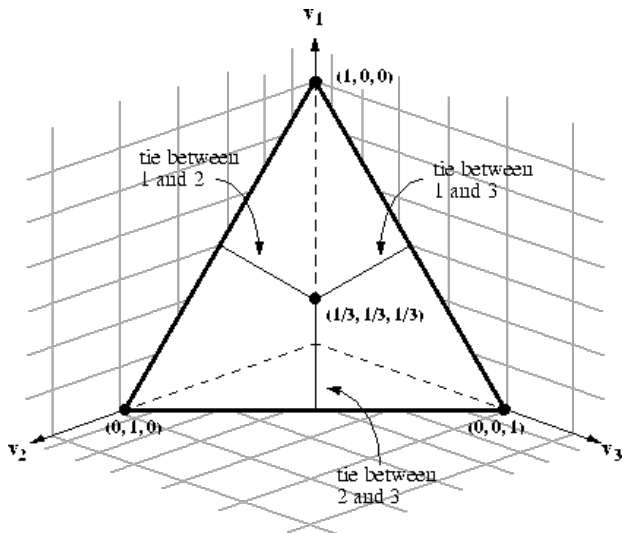
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- Note that for this example, the parameter space Θ is the $(K - 1)$ -*simplex* and has *no* interior in $\mathcal{R}^K$



Finding MLE via the Lagrangian method

- For this example, the problem of finding the MLE is given by the following *constrained* optimization problem:

$$\begin{aligned} &\text{Maximize} && \log p_{\theta}(x) \\ &\text{Subject to} && \sum_{k=1}^K p_k = 1 \\ & && p_k \geq 0, \quad \forall k \in [K] \end{aligned}$$

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- In optimization theory, the standard approach for solving constrained optimization problems are the *Lagrangian method*
- More specifically, let

$$L_{\lambda}(\theta) := \log p_{\theta}(x) - \lambda \sum_{k=1}^K p_k$$

where λ is the *Lagrangian multiplier*

- The MLE can then be found by solving the equation

$$\nabla_{\theta} L_{\lambda}(\theta) = 0$$

and then finding a Lagrangian multiplier $\lambda \geq 0$ to satisfy the constraint

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gives

$$p_k = \frac{n_k}{\lambda}, \quad \forall k \in [K]$$

- Finally, to satisfy the constraint

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and hence

$$\lambda = \sum_{k=1}^K n_k = n$$

- We thus conclude that the MLE for θ is given by the *empirical frequency* estimator

$$\hat{\theta} = \left(\frac{N_1}{n}, \dots, \frac{N_K}{n} \right)$$

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- Thus by the *Lehmann–Scheffé theorem*, for this example, the MLE is also the (unique) UMVUE

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- In the next lecture, we will discuss a method known as *expectation-maximization (EM)* algorithm, which is specifically designed for computing MLE for models with *latent* variables

Estimating $g(\theta)$

- Let $\eta = g(\theta)$. Note that if the likelihood function $f_{\theta}(x)$ can be *re-parameterized* in terms of η , then it is *natural* to use an *MLE* of η as an “*MLE*” estimate of $g(\theta)$

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- By the *re-parameterization* property, η^* is an *MLE* of η if and only if

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- Many other estimators such as UMVUE do *not* enjoy the invariance property